

VISUALIZING DIFFICULTY IN SIGHT-SINGING PRACTICE WITH COLOR-CODED SCORES

Hiroki Karatsu

Tokyo University of the Arts/AIST
karatsu-hiroki@aist.go.jp

Satoru Fukayama

AIST
s.fukayama@aist.go.jp

ABSTRACT

Although sight-singing is a fundamental skill for all musicians, many struggle to acquire it effectively due to its inherent difficulty. In order to support sight-singing practice, this study proposes a color-coded musical score that visualizes the estimated difficulty of each note. The difficulty is predicted using a bi-directional LSTM model trained on performance data (SPED), which achieved a moderate correlation on test data. A preliminary application suggests that the resulting visualization is reasonably consistent with perceived difficulty, based on the author's evaluation.

1. INTRODUCTION

Sight-singing is the ability to sing a piece of written music accurately after a short preview time, without having practiced it beforehand. It has long been regarded as a fundamental skill for all musicians. Some previous research has also highlighted its importance and demonstrated its relevance not only to other areas of solfège but also to practical skills [1]. Although learners admit the value of sight-singing [2], it is difficult to acquire this skill, so that many learners struggle with it and develop anxiety or a lack of confidence [3].

According to the previous research, scanning music to find challenging spots appears to be a valid strategy [4]. However, a previous study has shown that some learners sight-sing without using any specific strategies [5]. This may indicate that while some can perform accurately without conscious planning, others struggle due to a lack of strategic preparation. For these learners, visually marking difficult spots on the score could help them engage more strategically during the preview time.

Therefore, this study proposes a new color-coded score for sight-singing with visualization of difficulty. This tool aims to help learners make better use of their preview time by highlighting musically difficult points or areas, thereby supporting strategic and focused practice.

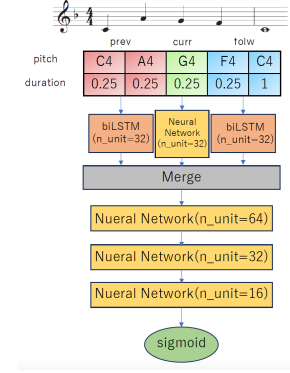


Figure 1. Estimation model architecture.

2. COLOR-CODED SCORE WITH DIFFICULTY INFORMATION

Previous study has explored the use of color-coded musical scores, reporting that such scores can enhance children's engagement [6]. In contrast to this prior approach, the method proposed in this study embeds musical information of difficulty directly into the colors themselves.

The difficulty of sight-singing can be defined subjectively by experts or objectively through statistical data. While traditional solfège education mainly relies on human judgment, data-driven approaches enable cost-effective, scalable evaluation and application into online education. This study adopts an objective definition based on the Sight-singing Performance Evaluation Dataset (SPED).

3. DIFFICULTY ESTIMATION MODEL

3.1 SPED

SPED is a dataset that contains a large number of sight-singing performance evaluations [7]. For each performance, every note is objectively labeled with a binary score (1 if correct, 0 otherwise) based on the accuracy of pitch and rhythm respectively. We adopted it as the following definition of difficulty and as a training/test dataset for estimator.

3.2 Definition of Difficulty

We define the difficulty of each note as the error rate across multiple performances. Specifically, the difficulty $\text{diff}_{n,k}$ of the k -th note in task n is defined as:



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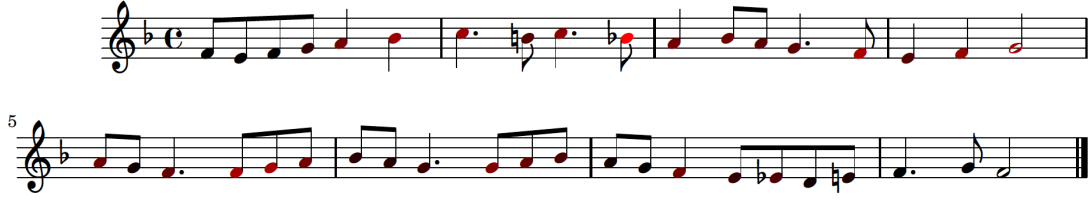


Figure 2. Color-coded score of the sample task 1.

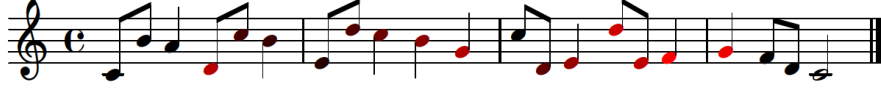


Figure 3. Color-coded score of the sample task 2.

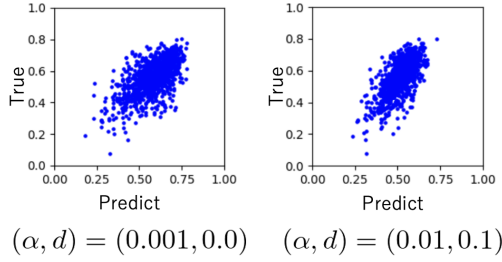


Figure 4. Scatter plots comparing predicted and true difficulty.

$$\text{diff}_{n,k} = 1 - \frac{\sum_{i=1}^{N_n} (p_{n,k,i} \cdot r_{n,k,i})}{N_n}$$

where N_n is the number of performances of task n , and $p_{n,k,i}$ and $r_{n,k,i}$ denote the pitch and rhythm correctness of the k -th note in the i -th performance. The product $p \cdot r$ represents whether both aspects were correct. To ensure consistency in performer proficiency and sufficient statistical reliability, only tasks with $N_n \geq 500$ were used from SPED.

3.3 Model Architecture

The difficulty of sight-singing a note depends on its musical context, such as intervals and rhythms with surrounding notes [8]. To capture this, the model uses a bi-directional LSTM layer followed by fully connected layers (Fig. 1). Each note is represented by one-hot vectors for pitch and duration. A similar architecture has proven to be effective in music information processing, e.g. DeepBach [9].

3.4 Training and Test Results

We trained the estimator for 100 epochs using input sequences of three preceding and succeeding notes. To evaluate performance, we tested dropout rates $d = 0.0, 0.1, 0.2, 0.3$ and learning rates $\alpha = 0.1, 0.01, 0.001$. The best result of correlation coefficient, $r = 0.689$, was obtained with $(\alpha, d) = (0.01, 0.1)$, while the lowest, $r = 0.616$, occurred at $(\alpha, d) = (0.001, 0.0)$. These scatter plots comparing predicted and ground-truth scores is

shown in Fig. 4.

4. GENERATING COLOR-CODED SCORES

Using the estimator, we generated scores that visualize note difficulty using a black-to-red gradient (from easy to difficult). Difficulty values were first normalized using min-max scaling within each task and then mapped to color. The first and last three notes were excluded from the evaluation and rests were treated as extensions of the preceding note. We applied this method to two sight-singing tasks from a Japanese textbook [10] (Fig. 2, 3).

4.1 Discussion of the Generated Scores

In the first task (Fig. 2), the model appropriately assigns high difficulty to the B $\flat$ in m.2, which follows a B $\natural$ —a semitone shift that can challenge novice learners. However, B $\natural$ itself is not rated as particularly difficult, despite being outside the key of F major; assigning it higher difficulty may be reasonable. Likewise, the chromatic sequence at the end of m.7 (E- E $\flat$ -D-E), could be typically difficult. In the second task (Fig. 3) which contains several leaps, some leaps were rated more challenging, while others received lower difficulty evaluations. Leaps of a minor seventh, such as at the end of m.3 (D-E-F), were more appropriately rated higher.

While the proposed model showed reasonable performance in some cases, further improvements are needed. We plan to adjust the context window length and explore Transformer-based architectures.

5. CONCLUSION

This study explored color-coded scores based on difficulty estimated using a bi-directional LSTM model. Although the model showed moderate accuracy, the model needs improvement before conducting user studies. We plan to conduct user studies with novice learners to assess the pedagogical validity of the estimated difficulty and its usefulness in sight-singing practice. A key challenge remains in identifying not only how difficult a note is but why—e.g., due to interval, rhythm, or chromaticism.

6. REFERENCES

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